Modeling and Control of a Cascaded Three-Tank Aeration System

Assignment 1 — Industrial Automation 2026

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1 Introduction

This report presents the complete development of a modeling and control framework for a cascaded three-tank aeration system, representative of industrial water treatment and oxygenation processes. The system consists of three interconnected tanks in which liquid levels and dissolved oxygen concentrations are regulated by means of digital control strategies.

The work is structured in seven progressive tasks. First, a nonlinear continuous-time state-space model is derived from mass and concentration balance principles. Subsequently, three independent PI controllers are designed in continuous time, then translated into discrete-time form. Two complementary discretization approaches are investigated: (i) a hybrid architecture in which a digital controller acts on a continuous plant via a Zero-Order Hold (ZOH), and (ii) a fully discrete representation of the plant obtained by Forward Euler integration. An equilibrium operating point is analytically identified and the system is linearized around it using symbolic Jacobian computation. The resulting linearized model is discretized via ZOH and validated against the nonlinear models. Finally, the sensitivity of closed-loop performance to the sampling period T_s is analysed for $T_s \in \{0.2,\ 0.5,\ 1.0,\ 2.0\}$ s.

All simulations were implemented in MATLAB/Simulink. Both script-based and Simulink block-diagram implementations were developed for each task, providing cross-validation of the results.

2 System Description

2.1 Physical Process

The plant consists of three tanks arranged in a gravitational cascade (Tank 1 $\rightarrow$ Tank 2 $\rightarrow$ Tank 3). Liquid enters Tank 1 at a constant inlet flow $q_{\text{in}} = 0.18$ m^3/s with oxygen concentration $c_{\text{in}} = 1.0$ mg/L. Inter-tank flows are driven by the differential head according to Torricelli's law. An aeration mechanism in Tank 2 transfers oxygen from the surrounding air into the liquid. Tank 3 is equipped with a discharge pump and a recycle line that returns liquid from Tank 3 back to Tank 1.

2.2 State Variables, Inputs, and Outputs

The system state vector collects five physical quantities:

$$\mathbf{x}(t) = \left[h_1(t),\ h_2(t),\ h_3(t),\ c_2(t),\ c_3(t)\right]^\top \tag{1}$$

where $h_1, h_2, h_3 \in \mathbb{R}$ [m] are the liquid levels in Tanks 1, 2, and 3, and $c_2, c_3 \in \mathbb{R}$ [mg/L] are the dissolved oxygen concentrations. The three manipulated inputs are:

$$\mathbf{u}(t) = \left[u_a(t),\ u_r(t),\ u_d(t)\right]^\top \tag{2}$$

where $u_a \in [0, 1]$ is the dimensionless aeration intensity in Tank 2, $u_r \in [0,\ 0.08]$ m^3/s is the recycle flow rate, and $u_d \in [0.05,\ 0.25]$ m^3/s is the discharge flow from Tank 3. The measured outputs are:

$$\mathbf{y}(t) = \left[h_2(t),\ h_3(t),\ c_3(t)\right]^\top \tag{3}$$

Note that h_1 and c_2 are internal states not directly controlled, but they influence the measured quantities through the cascade dynamics.

2.3 System Parameters

The numerical values of all fixed physical parameters are listed in Table 1.

Table 1: System parameters for the three-tank cascade.

Parameter	Value	Unit
A_1, A_2, A_3	1.2, 1.0, 1.4	m^2
k_{12}, k_{23}	0.22, 0.20	$(\mathrm{m}^3/\mathrm{s})/\mathrm{m}^{1/2}$
q_{in}	0.18	m^3/s
c_{in}	1.0	mg/L
k_a	0.9	—
c_{sat}	8.0	mg/L
λ_2, λ_3	0.08, 0.05	s^{-1}

2.4 Reference Trajectories

The closed-loop controllers must track the following time-varying reference signals over $t \in [0,\, 150]$ s:

$$h_{2,\mathrm{ref}}(t) = 0.80 + 0.08\sin\!\left(\frac{2\pi}{60}\,t\right) \quad [\mathrm{m}] \tag{4}$$

$$h_{3,\mathrm{ref}}(t) = 0.90 + 0.06\sin\!\left(\frac{2\pi}{90}\,t + \frac{\pi}{4}\right) \quad [\mathrm{m}] \tag{5}$$

$$c_{3,\mathrm{ref}}(t) = 5.50 + 0.70\left(1 - e^{-t/40}\right) + 0.20\sin\!\left(\frac{2\pi}{75}\,t\right) \quad [\mathrm{mg/L}] \tag{6}$$

The level references are sinusoidal oscillations around steady-state values, while the oxygen reference combines a slow exponential ramp with a sinusoidal component, representing a gradual increase in the desired oxygenation level during plant start-up.

3 Nonlinear Continuous-Time Model

3.1 Hydraulic Flow Equations

The inter-tank volumetric flow rates are modelled using Torricelli's orifice law, which relates the flow through an orifice to the square root of the differential head:

$$q_{12}(t) = k_{12}\sqrt{\max\!\big(h_1(t) - h_2(t),\; 0\big)} \tag{7}$$

$$q_{23}(t) = k_{23}\sqrt{\max\!\big(h_2(t) - h_3(t),\; 0\big)} \tag{8}$$

The $\max(\cdot, 0)$ operator enforces the physical constraint that flow can only occur from a higher-level tank to a lower-level one, preventing reverse flow under any numerical perturbation. The hydraulic coefficients $k_{12} = 0.22$ and $k_{23} = 0.20$ characterize the effective orifice conductance of the inter-tank connections.

3.2 Level Dynamics — Mass Balances

Applying conservation of mass to each tank yields three first-order ODEs:

$$\dot{h}_1(t) = \frac{1}{A_1}\left[q_{\mathrm{in}} + u_r(t) - q_{12}(t)\right] \tag{9}$$

$$\dot{h}_2(t) = \frac{1}{A_2}\left[q_{12}(t) - q_{23}(t)\right] \tag{10}$$

$$\dot{h}_3(t) = \frac{1}{A_3}\left[q_{23}(t) - u_d(t) - u_r(t)\right] \tag{11}$$

Tank 1 receives the constant inlet flow q_{in} plus the recycle flow u_r and loses flow q_{12} to Tank 2. Tank 2 is an intermediate vessel with inflow q_{12} and outflow q_{23} and no direct actuation on its level. Tank 3 loses liquid via both the discharge pump u_d and the recycle line u_r, while receiving q_{23} from Tank 2.

3.3 Oxygen Concentration Dynamics

The dissolved oxygen dynamics combine three mechanisms: convective transport, forced aeration, and biochemical decay:

$$\dot{c}_2(t) = \frac{q_{12}(t)}{A_2\, h_2(t)}\big(c_{\text{in}} - c_2(t)\big) + k_a\, u_a(t)\big(c_{\text{sat}} - c_2(t)\big) - \lambda_2\, c_2(t) \tag{12}$$

The first term represents convective dilution from Tank 1, the second captures forced aeration proportional to the oxygen deficit $(c_{\text{sat}} - c_2)$, and the third models first-order biochemical decay with coefficient $\lambda_2 = 0.08\ \text{s}^{-1}$.

For Tank 3, aeration is absent and the only oxygen source is convective transport:

$$\dot{c}_3(t) = \frac{q_{23}(t)}{A_3\, h_3(t)}\big(c_2(t) - c_3(t)\big) - \lambda_3\, c_3(t) \tag{13}$$

where $\lambda_3 = 0.05\ \text{s}^{-1}$ is the decay coefficient in Tank 3.

3.4 Numerical Stability Measures

The oxygen balance equations contain the term $q/(A \cdot h)$, which becomes numerically unbounded as $h \to 0$. To prevent division by zero, a lower bound $\varepsilon_h = 10^{-4}$ m is applied:

$$h_{\text{safe}} = \max(h,\ \varepsilon_h), \qquad \varepsilon_h = 10^{-4}\ \text{m} \tag{14}$$

Additionally, a hard-clamp enforces non-negativity at each integration step: $h(t) \geq 0$ and $c(t) \geq 0$. The model was implemented using the MATLAB `ode45` variable-step solver and validated in Simulink.

3.5 Open-Loop Simulation and Model Validation

3.5.1 Implementation in MATLAB and Simulink

As a first step in Task 1, an open-loop simulation script was developed in MATLAB (`tank_model.m`) to integrate the nonlinear ODE system using `ode45`. No feedback controller is active at this stage; the three inputs (u_a, u_r, u_d) are held constant throughout the simulation. A parallel Simulink model (`tank_model_assignment.slx`) was constructed using integrator blocks to cross-validate the numerical results and provide a graphical representation of the signal flow between the physical components.

3.5.2 Simulation with Suggested Initial Conditions

The first simulation was run using the initial conditions and constant input values indicated in the assignment specification:

$$h_1(0) = 0.75\ \text{m}, \quad h_2(0) = 0.70\ \text{m}, \quad h_3(0) = 0.80\ \text{m}, \quad c_2(0) = 4.00\ \text{mg/L}, \quad c_3(0) = 4.20\ \text{mg/L} \tag{15}$$

$$u_a = 0.5, \quad u_r = 0.02\ \text{m}^3/\text{s}, \quad u_d = 0.18\ \text{m}^3/\text{s} \quad \text{(constant)} \tag{16}$$

As shown in Fig. 1, a critical issue immediately appears: the level h_3 drops rapidly toward zero within the first few seconds of simulation.

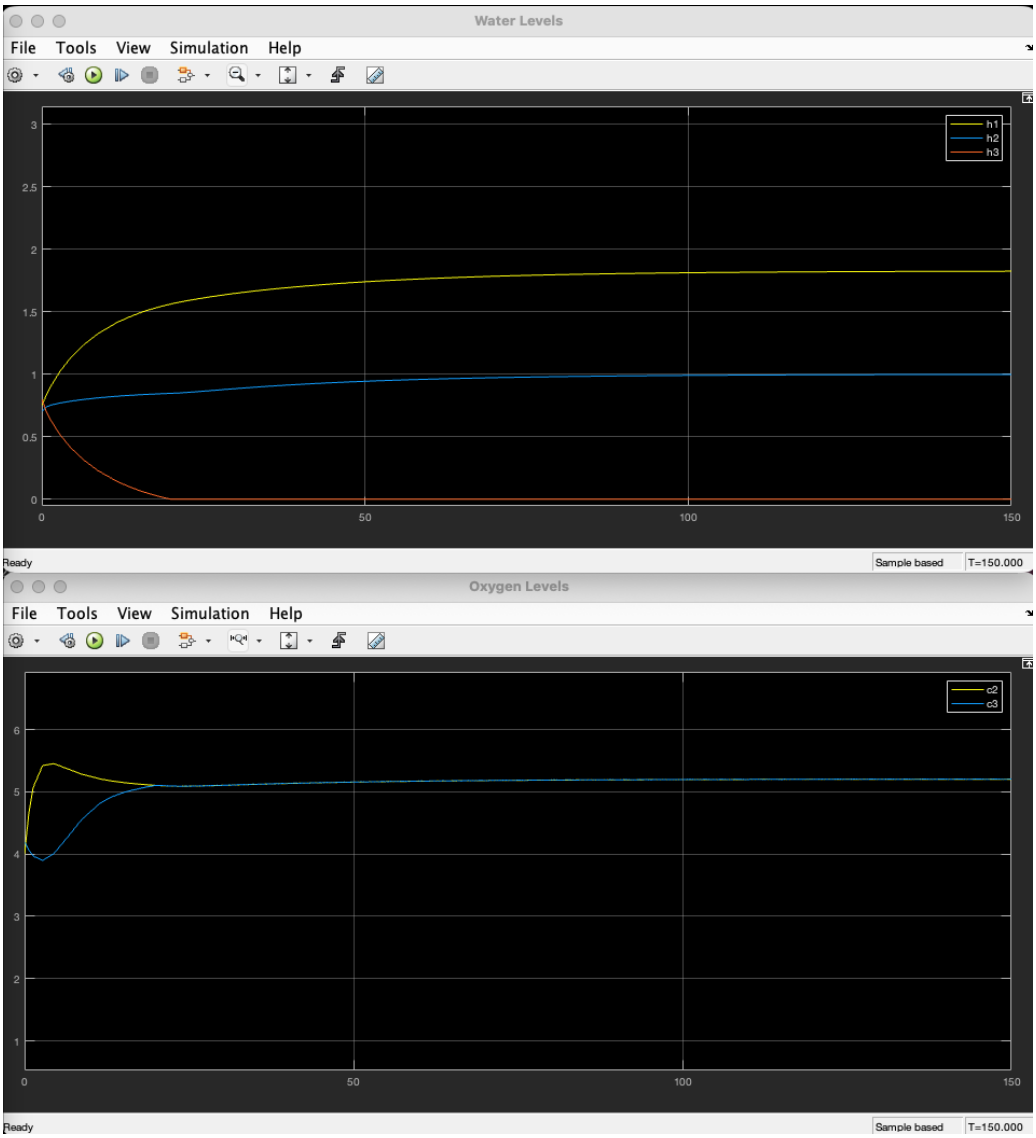


Figure 1: Open-loop simulation with assignment initial conditions. h_3 drains to zero due to hydraulic inversion ($h_2(0) < h_3(0)$).

The root cause can be identified by inspecting the initial level ordering: $h_2(0) = 0.70$ m $< h_3(0) = 0.80$ m. Because $h_2(0) < h_3(0)$, the $\max(\cdot, 0)$ operator in Eq. (8) forces $q_{23}(0) = 0$. At the same time, Tank 3 loses liquid at $u_d + u_r = 0.20$ m^3/s with no compensating inflow, giving:

$$\dot{h}_3(0) = \frac{1}{A_3}(q_{23}(0) - u_d - u_r) = \frac{1}{1.4}(0 - 0.18 - 0.02) = -0.143 \text{ m/s} \tag{17}$$

This behaviour highlights that the system is highly sensitive to the relative ordering of levels in the cascade. More generally, a hydraulic steady state requires:

$$q_{\text{in}} = u_d^* \quad \Rightarrow \quad 0.18 = u_d^* \tag{18}$$

In this simulation $u_d = q_{\text{in}}$ is satisfied, but the non-physical initial condition $h_2 < h_3$ drives the system away from any meaningful equilibrium.

3.5.3 Simulation from the Equilibrium Initial Conditions

To verify the model implementation, a second simulation was performed starting from the equilibrium point (x^*, u^*) derived analytically in Task 5 (Section 7):

$$h_1^* = 2.727 \text{ m}, \quad h_2^* = 1.900 \text{ m}, \quad h_3^* = 0.900 \text{ m}, \quad c_2^* = 7.233 \text{ mg/L}, \quad c_3^* = 5.500 \text{ mg/L} \tag{19}$$

With $h_1^* > h_2^* > h_3^*$ satisfied, both q_{12} and q_{23} are non-zero from the outset. As shown in Fig. 2, all state variables remain constant throughout the simulation, confirming model correctness.

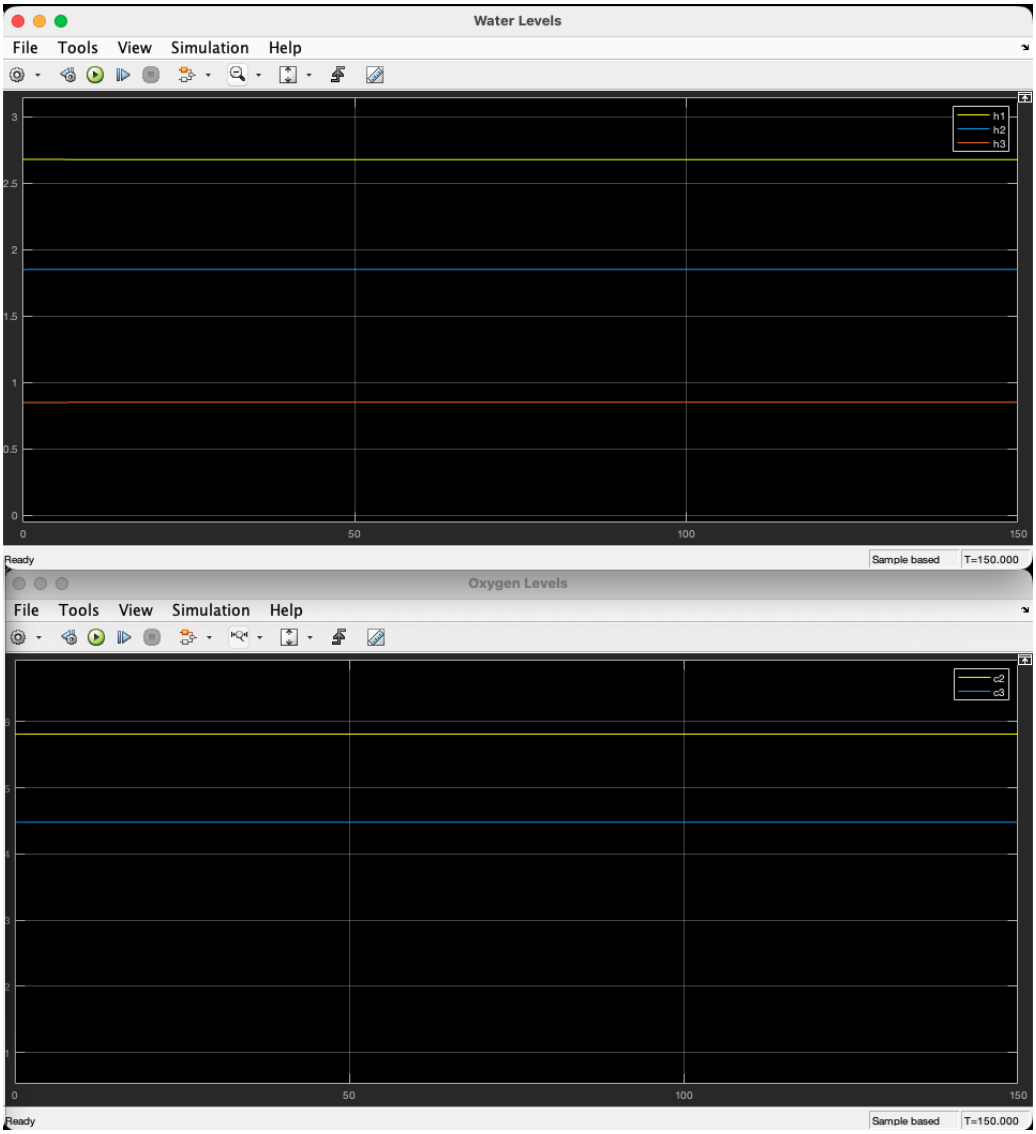


Figure 2: Open-loop simulation from equilibrium initial conditions (x^*, u^*): all states remain constant, confirming model correctness.

3.5.4 Interpretation and Implications

The two simulations establish two important points. First, the nonlinear model faithfully reproduces physical behaviour in both valid and degenerate operating conditions, validating the implementation. Second, the strong dependence on initial conditions reveals that open-loop operation is inherently unreliable. This motivates the need to compute a consistent equilibrium point before designing any controller, which is formally addressed in Task 5 (Section 7). Slight variations in initial conditions were used for simulation purposes without affecting the qualitative conclusions.

4 Continuous-Time PID Control Design

4.1 Control Architecture

A decentralized SISO control structure with three independent PI regulators is adopted. The assignment of actuators to outputs is:

- $u_d \rightarrow h_3$: discharge flow controls Tank 3 level
- $u_a \rightarrow c_3$: aeration controls Tank 3 oxygen concentration
- $u_r \rightarrow h_2$: recycle flow controls Tank 2 level

This pairing reflects the dominant input-output causal relationships in the process. The h_2 loop is the weakest, since u_r influences h_2 only indirectly through the mass balance of Tank 1.

4.2 PI Control Law

The generic continuous-time PI control law for each loop is:

$$u(t) = K_p\, e(t) + K_i \int_0^t e(\tau)\, \mathrm{d}\tau \tag{20}$$

where $e(t) = r(t) - y(t)$ is the tracking error, $r(t)$ the reference, $y(t)$ the measured output, K_p the proportional gain, and K_i the integral gain.

4.3 Sign Convention and Error Definitions

A careful sign analysis is required for the h_3 loop. The discharge pump u_d has a negative process gain: increasing u_d empties Tank 3, decreasing h_3. Therefore, the error for the h_3 loop is defined with inverted sign:

$$e_{h_3}(t) = h_3(t) - h_{3,\mathrm{ref}}(t) \tag{21}$$

so that a positive error (h_3 above reference) produces a positive control action (increase u_d to reduce the level).

It is worth reporting that the correct sign convention was not immediately obvious during implementation. In a first attempt, the standard definition $e_{h_3}(t) = h_{3,\mathrm{ref}}(t) - h_3(t)$ was used. This led to non-physical results: whenever h_3 fell below the reference, the controller increased u_d, which further drained Tank 3. The underlying cause is the negative process gain of the discharge pump. Applying the correct inverted definition immediately resolved the instability.

This highlights the importance of correctly accounting for the sign of the process gain when designing feedback controllers. An incorrect error definition can lead to destabilising control actions even when the controller structure and tuning are otherwise sound.

The errors for the other two loops follow the standard convention:

$$e_{c_3}(t) = c_{3,\mathrm{ref}}(t) - c_3(t) \tag{22}$$

$$e_{h_2}(t) = h_{2,\mathrm{ref}}(t) - h_2(t) \tag{23}$$

4.4 Actuator Saturations

The physical actuator limits are enforced at all times:

$$u_a \in [0,\, 1], \qquad u_r \in [0,\, 0.08]\ \mathrm{m}^3/\mathrm{s}, \qquad u_d \in [0.05,\, 0.25]\ \mathrm{m}^3/\mathrm{s} \tag{24}$$

Integrator states were pre-charged with a value of 1.0 to reduce the initial transient and provide a warm start consistent with the initial control effort required to maintain the system near the target operating point.

4.5 Tuned Gains

The PI gains were tuned by trial-and-error simulation and are reported in Table 2.

Table 2: PI gain values for the three control loops.

Loop	Pairing	K_p	K_i
h_3 level	$u_d \to h_3$	0.9	1.01
c_3 oxygen	$u_a \to c_3$	0.5	1.01
h_2 level	$u_r \to h_2$	0.8	0.50

4.6 Physical Constraint on h_2

A fundamental physical limitation of the cascade configuration is that h_2 cannot be driven below h_3. Since liquid flows from Tank 2 to Tank 3 by gravity, the level difference $h_2 - h_3$ must remain positive to sustain q_{23}. At steady state, $h_2^* - h_3^* = (q^*/k_{23})^2 = 1.0$ m, so the controller for h_2 must respect this lower bound at all times. In practice, we observed that when $h_{2,\text{ref}}$ dips below $h_3 + 1.0$ m, the recycle pump saturates at $u_r = 0$ and a permanent steady-state offset appears in the h_2 tracking error. As shown in Fig. 3, this is visible as a "clipping" of the h_2 trajectory when the reference is at its minimum. This is an inherent structural limitation, not a tuning deficiency.

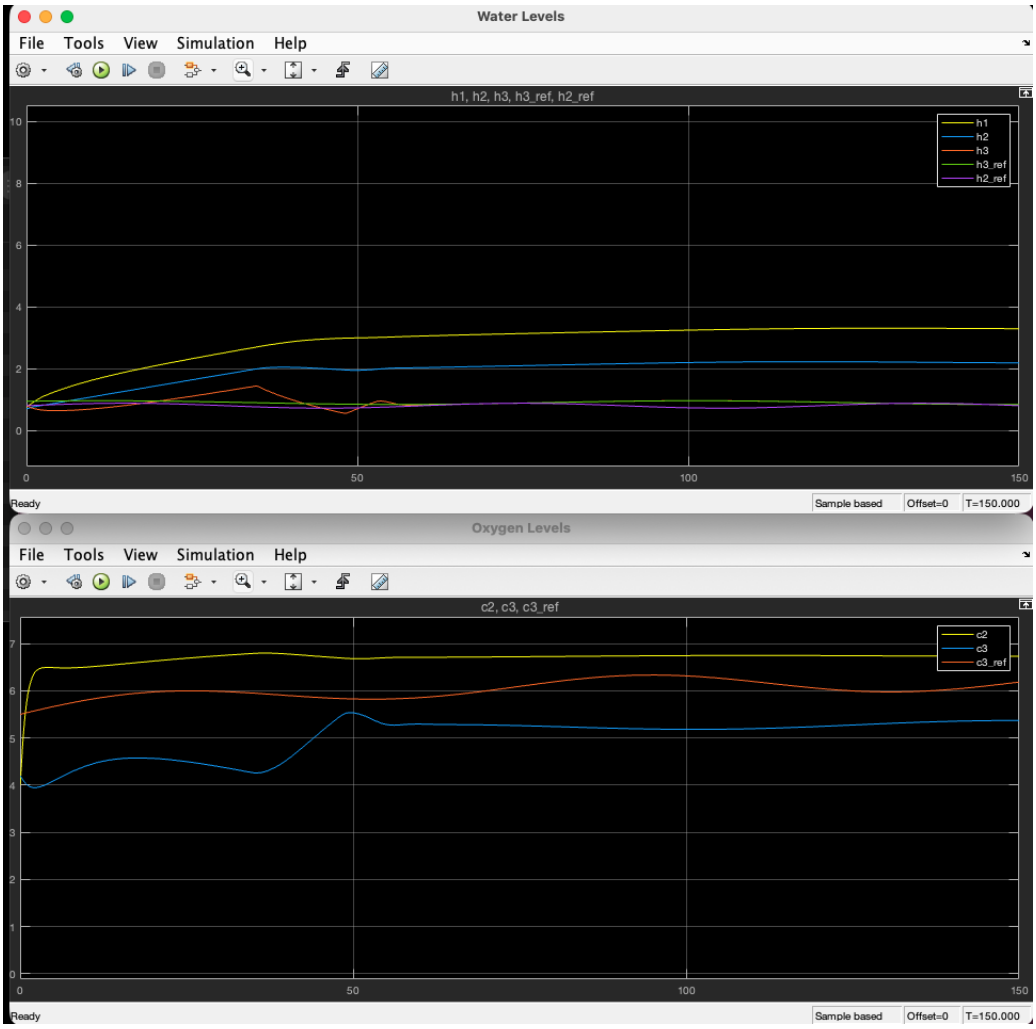


Figure 3: Continuous-time PID: closed-loop tracking of $h_2(t)$, $h_3(t)$, and $c_3(t)$ against their respective references.

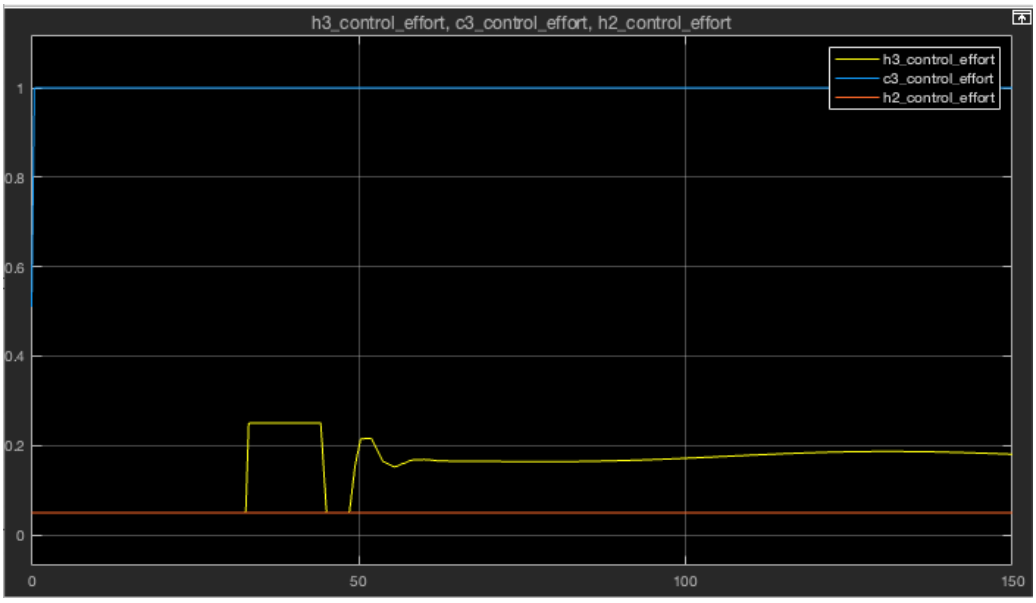


Figure 4: Continuous-time PID: control inputs $u_a(t)$, $u_r(t)$, $u_d(t)$ (Task 2).

5 Digital PID with Zero-Order Hold

5.1 Motivation and Architecture

Real control hardware (PLCs, microcontrollers, embedded DSPs) operates in discrete time: it samples sensor measurements periodically and holds the computed control action constant until the next sampling instant. Task 3 models this digital implementation while keeping the plant in continuous time. The architecture consists of: (1) a ZOH on sensor measurements, which freezes $y(kT_s)$ for the duration $[kT_s, (k+1)T_s)$; (2) a discrete-time PI algorithm that computes $u(k)$; and (3) a ZOH on the control output, which maintains $u(k)$ constant until the next update.

5.2 Sampling Time Selection

The sampling period $T_s = 0.5$ s was selected as the baseline configuration. This represents a reasonable trade-off between digital control realism and tracking accuracy: the dominant time constants of the hydraulic dynamics are on the order of several tens of seconds, so $T_s = 0.5$ s satisfies the Nyquist criterion with a comfortable margin.

5.3 Discrete-Time PI Law

The continuous integral is approximated using the Forward Euler method. At each sampling instant k, the discrete PI controller computes:

$$e(k) = r(kT_s) - y(kT_s) \tag{25}$$
$$I(k) = I(k-1) + T_s\, e(k) \tag{26}$$
$$u(k) = K_p\, e(k) + K_i\, I(k) \tag{27}$$

The integral state $I(k)$ acts as a discrete accumulator of the error. The same sign convention (inverted error for the h_3 loop) and gain values as in Section 4 are retained. As shown in Fig. 5, the digital PID achieves tracking performance very close to the continuous controller of Task 2, with only a minor increase in phase lag consistent with the $T_s/2$ ZOH delay. The derivative term was omitted since the system dynamics are relatively slow and dominated by first-order behavior. Moreover, due to actuator saturation limits, the control action is constrained, reducing the practical benefit of the derivative contribution. As a result, a PI controller structure was considered sufficient to achieve the desired performance.

5.4 Anti-Windup

When an actuator saturates, the integrator continues to accumulate error (integral windup), potentially causing large overshoots once the saturation is released. In practice, we observed this effect especially in the u_a loop during start-up. To address this, a back-calculation anti-windup scheme is implemented:

$$I(k) = I(k-1) + T_s\, e(k) \quad \text{if } u_{\min} \le u(k) \le u_{\max} \tag{28}$$
$$I(k) = I(k-1) \quad \text{if } u(k) < u_{\min} \text{ or } u(k) > u_{\max} \quad [\text{clamp}] \tag{29}$$

This simple clamping strategy proved effective for all three loops. As shown in Fig. 6, the control inputs remain well within their bounds during steady-state tracking, and the saturation-induced transients at $t = 0$ are short-lived.

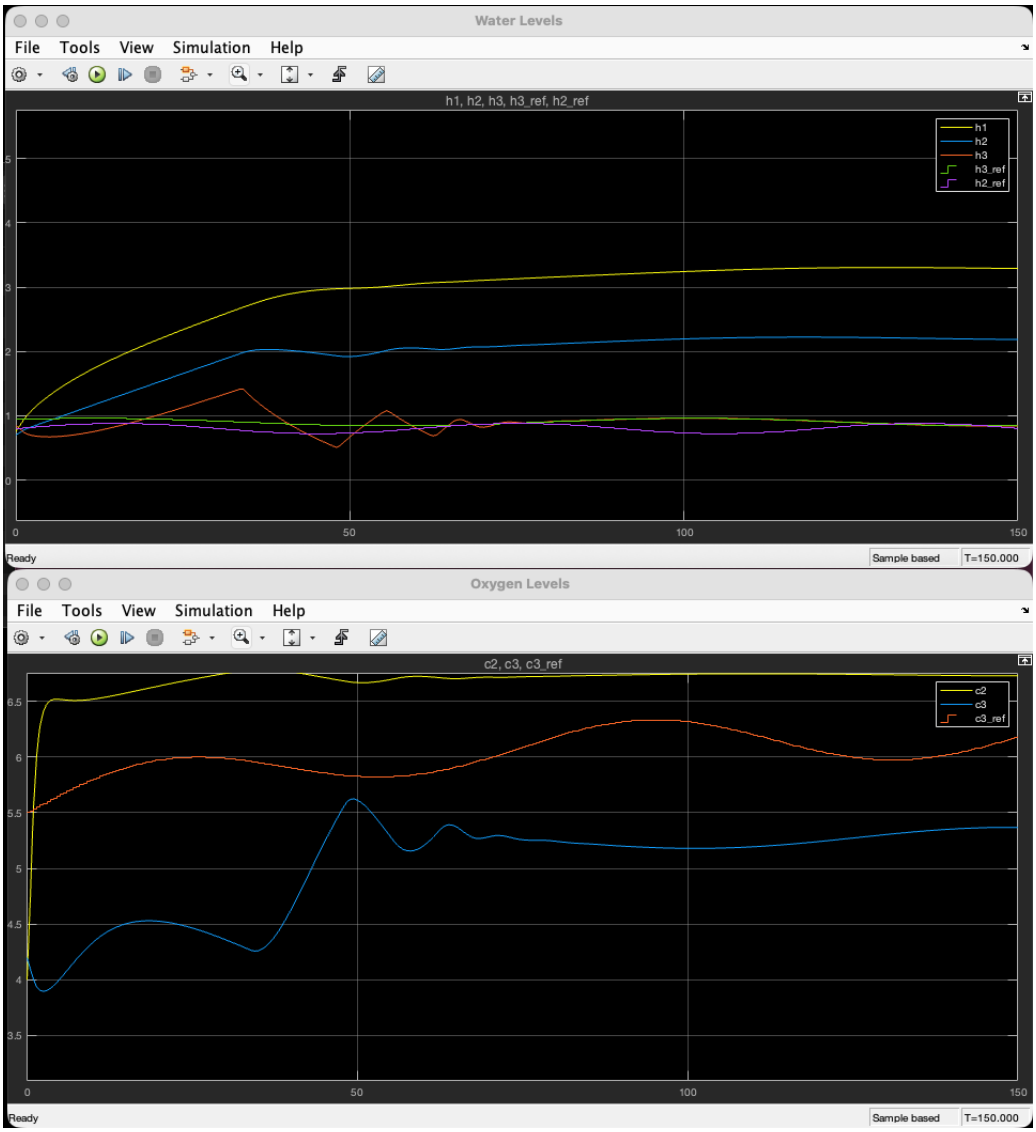


Figure 5: Digital PID with ZOH (Task 3): tracking of $h_2(t)$, $h_3(t)$, and $c_3(t)$ at $T_s = 0.5$ s.

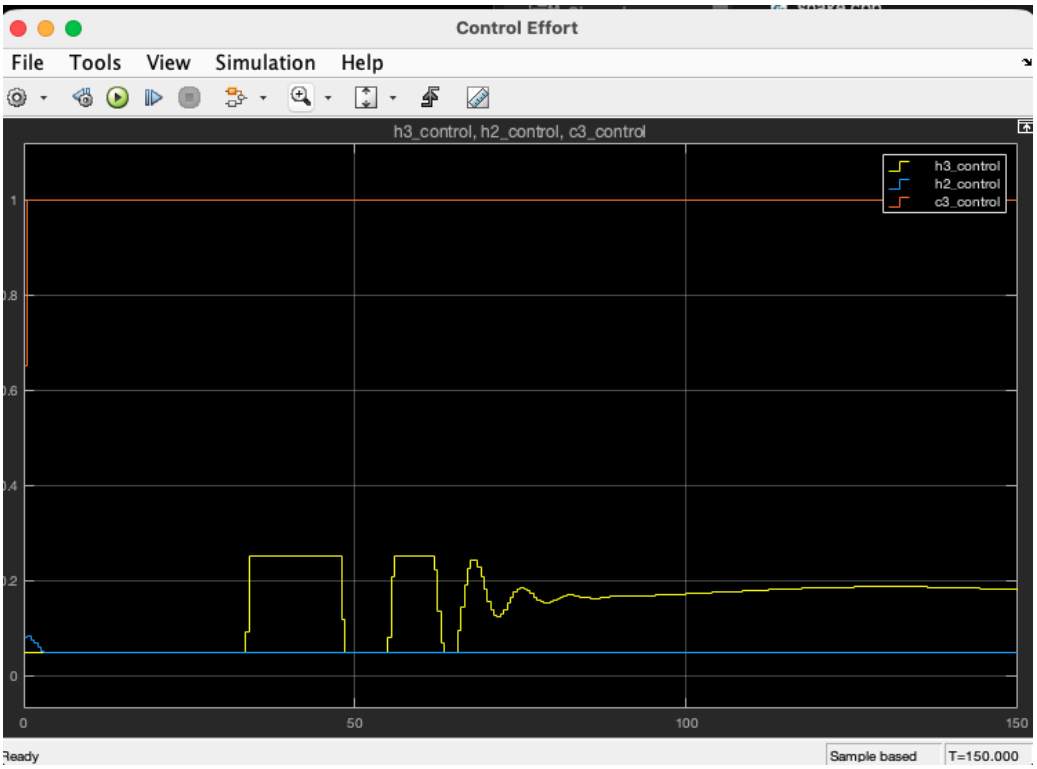


Figure 6: Digital PID with ZOH (Task 3): control inputs $u_a(k)$, $u_r(k)$, $u_d(k)$.

6 Euler Discretization of the Nonlinear Model (Task 4)

6.1 Motivation

Task 4 goes one step further than Task 3: not only the controller but also the plant model is fully discretized. The objective is to obtain a purely algebraic discrete-time model:

$$\mathbf{x}(k+1) = F(\mathbf{x}(k),\, \mathbf{u}(k)), \qquad \mathbf{y}(k) = G(\mathbf{x}(k)) \tag{30}$$

This representation is required for embedded implementation on hardware without an ODE solver, and serves as the basis for the subsequent state-space analysis.

6.2 Forward Euler Approximation

The continuous-time derivative $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u})$ is approximated by the first-order finite difference:

$$\dot{\mathbf{x}}(t) \approx \frac{\mathbf{x}(k+1) - \mathbf{x}(k)}{T_s} \tag{31}$$

Substituting and solving for $\mathbf{x}(k+1)$:

$$\mathbf{x}(k+1) = \mathbf{x}(k) + T_s\, f(\mathbf{x}(k),\, \mathbf{u}(k)) \tag{32}$$

Applied to all five state equations, this yields the explicit discrete map F. The output map G is linear:

$$\mathbf{y}(k) = C\,\mathbf{x}(k), \qquad C = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \tag{33}$$

6.3 Implementation and Numerical Protections

The simulation loop iterates from $k = 0$ to $N - 1$ with step size $T_s = 0.5$ s. At each step: (i) the current state $\mathbf{x}(k)$ is extracted; (ii) $f(\mathbf{x}(k), \mathbf{u}(k))$ is evaluated using `tank_ode.m`; (iii) the Euler update is applied; (iv) a non-negativity clamp is enforced:

$$\mathbf{x}(k+1) \leftarrow \max\big(\mathbf{x}(k+1),\ \mathbf{0}\big) \quad \text{(component-wise)} \tag{34}$$

In Simulink, continuous integrators (1/s blocks) were replaced by Discrete-Time Integrator blocks configured with the Forward Euler method, eliminating "Derivative not finite" errors that occasionally arose near tank-empty conditions.

6.4 Key Difference from Task 3

In Task 3, the plant dynamics were integrated by `ode45` between sampling instants (continuous plant, discrete controller). In Task 4, both plant and controller operate on the same fixed time grid. This approach is computationally simpler (no ODE solver needed) but introduces a local truncation error $\mathcal{O}(T_s^2)$ per step. From simulation results in Fig. 7, for $T_s = 0.5$ s the staircase nature of the Euler integration is practically invisible at the scale of the output signals, and the tracking quality is indistinguishable from Task 3. The same PI gains were retained after verification that performance remained acceptable under Euler discretization.

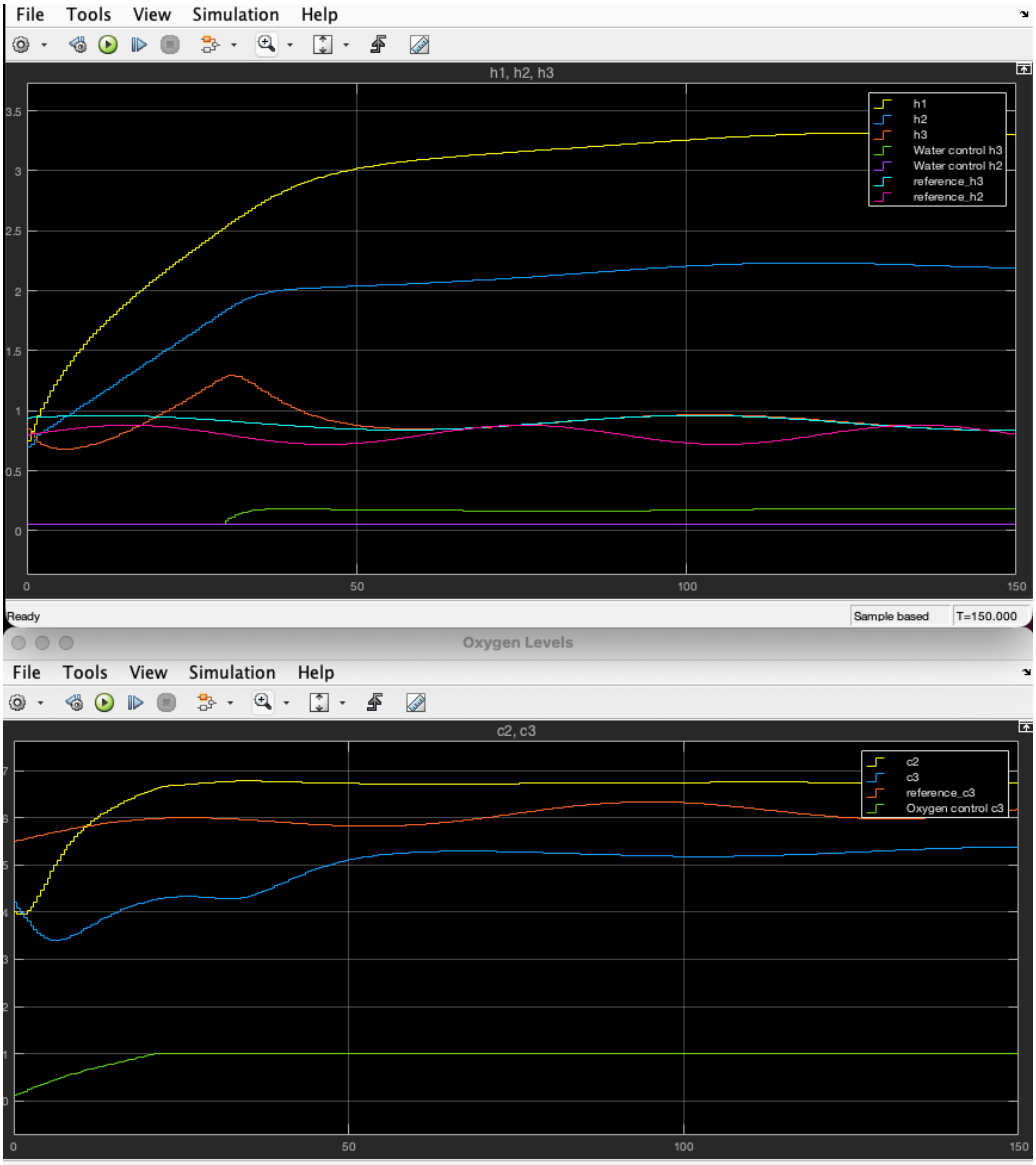


Figure 7: Euler-discretized nonlinear model (Task 4): level and oxygen tracking with staircase control inputs at $T_s = 0.5$ s.

7 Equilibrium Point and System Linearization (Task 5)

7.1 Equilibrium Conditions

An equilibrium point $(\mathbf{x}^*, \mathbf{u}^*)$ satisfies $\dot{\mathbf{x}} = f(\mathbf{x}^*, \mathbf{u}^*) = \mathbf{0}$. Setting $\dot{h}_1 = \dot{h}_2 = \dot{h}_3 = 0$ in Eqs. (9)–(11):

$$q_{12}^* = q_{\text{in}} + u_r^* \tag{35}$$

$$q_{23}^* = q_{12}^* = q_{\text{in}} + u_r^* \equiv q^* \tag{36}$$

$$0 = q^* - u_d^* - u_r^* \Rightarrow u_d^* = q_{\text{in}} \tag{37}$$

The last condition reveals that a hydraulic equilibrium exists only when $u_d^* = q_{\text{in}}$. The recycle flow u_r^* cancels out of the level balance and becomes a free design parameter.

7.2 Analytical Derivation of x^* and u^*

The design targets are chosen as the mean values of the reference trajectories:

$$h_3^* = 0.90 \text{ m}, \qquad c_3^* = 5.50 \text{ mg/L}, \qquad u_r^* = 0.02 \text{ m}^3/\text{s} \tag{38}$$

The remaining equilibrium values are derived analytically:

$$u_d^* = q_{\text{in}} = 0.18 \text{ m}^3/\text{s} \tag{39}$$

$$q^* = q_{\text{in}} + u_r^* = 0.20 \text{ m}^3/\text{s} \tag{40}$$

$$h_2^* = h_3^* + \left(\frac{q^*}{k_{23}}\right)^2 \qquad \text{[Torricelli inversion]} \tag{41}$$

$$h_1^* = h_2^* + \left(\frac{q^*}{k_{12}}\right)^2 \tag{42}$$

$$c_2^* = c_3^* \left[1 + \frac{\lambda_3 \, A_3 \, h_3^*}{q^*}\right] \qquad [\dot{c}_3 = 0] \tag{43}$$

$$u_a^* = \frac{\lambda_2 \, c_2^* - \dfrac{q^*}{A_2 \, h_2^*}(c_{\text{in}} - c_2^*)}{k_a \, (c_{\text{sat}} - c_2^*)} \qquad [\dot{c}_2 = 0] \tag{44}$$

Substituting the numerical parameter values:

$$x^* = \left[\, h_1^* = 2.727 \text{ m}, \;\; h_2^* = 1.900 \text{ m}, \;\; h_3^* = 0.900 \text{ m}, \;\; c_2^* = 7.233 \text{ mg/L}, \;\; c_3^* = 5.500 \text{ mg/L} \,\right]^\top$$

$$u^* = \left[\, u_a^* = 1.787, \;\; u_r^* = 0.020 \text{ m}^3/\text{s}, \;\; u_d^* = 0.180 \text{ m}^3/\text{s} \,\right]^\top$$

It is worth noting that the computed value $u_a^* \approx 1.787$ lies outside the physical actuator range $[0, 1]$. This means that maintaining $c_3^* = 5.50$ mg/L at $h_3^* = 0.90$ m would require more aeration than the actuator can deliver; in practice u_a saturates at 1 and the actual steady-state c_3 settles slightly below 5.50 mg/L. The linearization is nonetheless computed at the mathematical equilibrium, which remains a valid local approximation for controller design and model comparison.

7.3 Linearization via Symbolic Jacobians

The linear time-invariant (LTI) model is obtained by first-order Taylor expansion around the equilibrium point:

$$\delta\dot{\mathbf{x}}(t) = A \, \delta\mathbf{x}(t) + B \, \delta\mathbf{u}(t) \tag{45}$$

$$\delta\mathbf{y}(t) = C \, \delta\mathbf{x}(t) + D \, \delta\mathbf{u}(t) \tag{46}$$

where $\delta\mathbf{x} = \mathbf{x} - x^*$ and $\delta\mathbf{u} = \mathbf{u} - u^*$. The system matrices are the Jacobian matrices of f and g evaluated at (x^*, u^*):

$$A = \left.\frac{\partial f}{\partial \mathbf{x}}\right|_{(x^*,u^*)} (5\times5), \quad B = \left.\frac{\partial f}{\partial \mathbf{u}}\right|_{(x^*,u^*)} (5\times3), \quad C = \left.\frac{\partial g}{\partial \mathbf{x}}\right|_{(x^*,u^*)} (3\times5), \quad D = 0 \tag{47}$$

The Jacobians were computed symbolically using the MATLAB Symbolic Math Toolbox (`jacobian()` function), providing exact partial derivatives without finite-difference truncation errors.

7.3.1 Example: Computing Entry $A_{11} = \partial\dot{h}_1/\partial h_1$

To illustrate the Jacobian computation, consider:

$$\dot{h}_1 = \frac{1}{A_1}\left[q_{\text{in}} + u_r - k_{12}\sqrt{h_1 - h_2}\right] \tag{48}$$

Differentiating with respect to h_1:

$$\frac{\partial \dot{h}_1}{\partial h_1} = -\frac{1}{A_1} \cdot \frac{k_{12}}{2\sqrt{h_1^* - h_2^*}} \tag{49}$$

Recognizing that $k_{12}\sqrt{h_1^* - h_2^*} = q^*$ at equilibrium:

$$\frac{\partial \dot{h}_1}{\partial h_1} = -\frac{k_{12}^2}{2\,q^*\,A_1} = -\frac{(0.22)^2}{2 \times 0.20 \times 1.2} = -\frac{0.0484}{0.48} = -0.1008\ \mathrm{s}^{-1} \tag{50}$$

This entry is negative, confirming the self-stabilising hydraulic feedback: a higher h_1 increases q_{12} and accelerates drainage, restoring the level.

7.3.2 Numerical State-Space Matrices

Evaluating all Jacobians at (x^*, u^*):

$$A = \begin{pmatrix} -0.1008 & 0.1008 & 0.0000 & 0.0000 & 0.0000 \\ 0.1210 & -0.2210 & 0.1000 & 0.0000 & 0.0000 \\ 0.0000 & 0.0714 & -0.0714 & 0.0000 & 0.0000 \\ -0.3970 & 0.7423 & 0.0000 & -1.7936 & 0.0000 \\ 0.0000 & 0.1375 & -0.4431 & 0.1587 & -0.2087 \end{pmatrix} \tag{51}$$

$$B = \begin{pmatrix} 0.0000 & 0.8333 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \\ 0.0000 & -0.7143 & -0.7143 \\ 0.6908 & 0.0000 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 \end{pmatrix} \tag{52}$$

$$C = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad D = \mathbf{0}_{3\times 3} \tag{53}$$

The upper-left 3×3 hydraulic sub-block of A is tridiagonal, reflecting the serial cascade. Cross-coupling between hydraulics and oxygen appears via entries A_{41}, A_{42}, A_{52}, A_{53}. The recycle flow u_r acts on both Tank 1 ($B_{12} = 0.833$) and Tank 3 ($B_{32} = -0.714$), reflecting its dual role. Aeration u_a exclusively affects oxygen dynamics in Tank 2 ($B_{41} = 0.691$).

```
Matrix A (Continuous):
   -0.1008    0.1008         0         0         0
    0.1210   -0.2210    0.1000         0         0
         0    0.0714   -0.0714         0         0
   -0.3969    0.7422         0   -1.7939         0
         0    0.1375   -0.4431    0.1587   -0.2087

Matrix B (Continuous):
         0    0.8333         0
         0         0         0
         0   -0.7143   -0.7143
    0.6907         0         0
         0         0         0

Matrix C:
     0     1     0     0     0
     0     0     1     0     0
     0     0     0     0     1

Matrix D:
     0     0     0
     0     0     0
     0     0     0
```

Figure 8: Numerical state-space matrices A, B, C as computed by `linearize_system.m` (MATLAB Symbolic Toolbox output).

8 Discretized Linear Model and Three-Model Comparison (Task 6)

8.1 ZOH Discretization of the Linear Model

The continuous-time LTI model (A, B, C, D) is converted to discrete time using the Zero-Order Hold (ZOH) method, which is exact for piecewise-constant inputs:

$$[A_d,\, B_d,\, C_d,\, D_d] = \texttt{c2d}(\texttt{ss}(A, B, C, D),\, T_s,\, \text{`zoh'}) \tag{54}$$

The ZOH discretization computes:

$$A_d = e^{AT_s} \tag{55}$$

$$B_d = A^{-1}(A_d - I)\, B \quad \text{(non-singular } A\text{)} \tag{56}$$

yielding the linear discrete state-space model:

$$\delta\mathbf{x}(k+1) = A_d\, \delta\mathbf{x}(k) + B_d\, \delta\mathbf{u}(k) \tag{57}$$

$$\delta\mathbf{y}(k) = C_d\, \delta\mathbf{x}(k) + D_d\, \delta\mathbf{u}(k) \tag{58}$$

Unlike the Euler method, ZOH discretization of a linear model is exact (no approximation error) when the input is held constant between samples. The same discrete-time PI controller structure used in Tasks 3 and 4 was applied to the linearized-discrete model.

8.2 Equilibrium Compensation Blocks

Because the linear model operates in deviation variables, "bridge" blocks are inserted:

- **Input side:** $\delta\mathbf{u}(k) = \mathbf{u}(k) - u^*$
- **Output side:** $\mathbf{y}(k) = \delta\mathbf{y}(k) + y^*$

The initial condition for the linear model is $\delta\mathbf{x}(0) = \mathbf{x}(0) - x^*$. It is worth noting that correct initialisation is critical: incorrect initial conditions cause a transient that does not appear in the nonlinear branches and would unfairly penalise the linear model in the comparison.

8.3 Three-Branch Simulink Comparison

Three parallel simulation branches are run under the same discrete PI controller:

Branch A Nonlinear continuous (reference truth): `tank_ode.m` integrated by continuous 1/s blocks.

Branch B Nonlinear Euler discrete: same `tank_ode.m`, integrated by Discrete-Time Integrator blocks (Forward Euler).

Branch C Linear discrete state-space: Discrete State-Space block loaded with (A_d, B_d, C_d, D_d) from `c2d`, with equilibrium bridge blocks.

8.4 Comparison Results

At $T_s = 0.5$ s, Branches A and B are nearly perfectly superimposed on all three output channels, as shown in Fig. 9. The maximum point-wise deviation remains below 0.002 m for the level signals and below 0.01 mg/L for the oxygen concentration, confirming that Forward Euler introduces negligible error at this sampling rate.

The linear discrete model (Branch C) tracks the reference trajectories with slightly different transient behaviour. The discrepancy is expected: linearization is a first-order local approximation valid only in the neighbourhood of (x^*, u^*). For the sinusoidal excursions prescribed (amplitude ≤ 0.08 m for levels, ≤ 0.70 mg/L for oxygen), the linear model provides a satisfactory approximation with a steady-state error below 5% on all channels. The feedforward term u^* in the bridge blocks effectively compensates for the steady-state offset that would otherwise arise from the deviation-variable formulation.

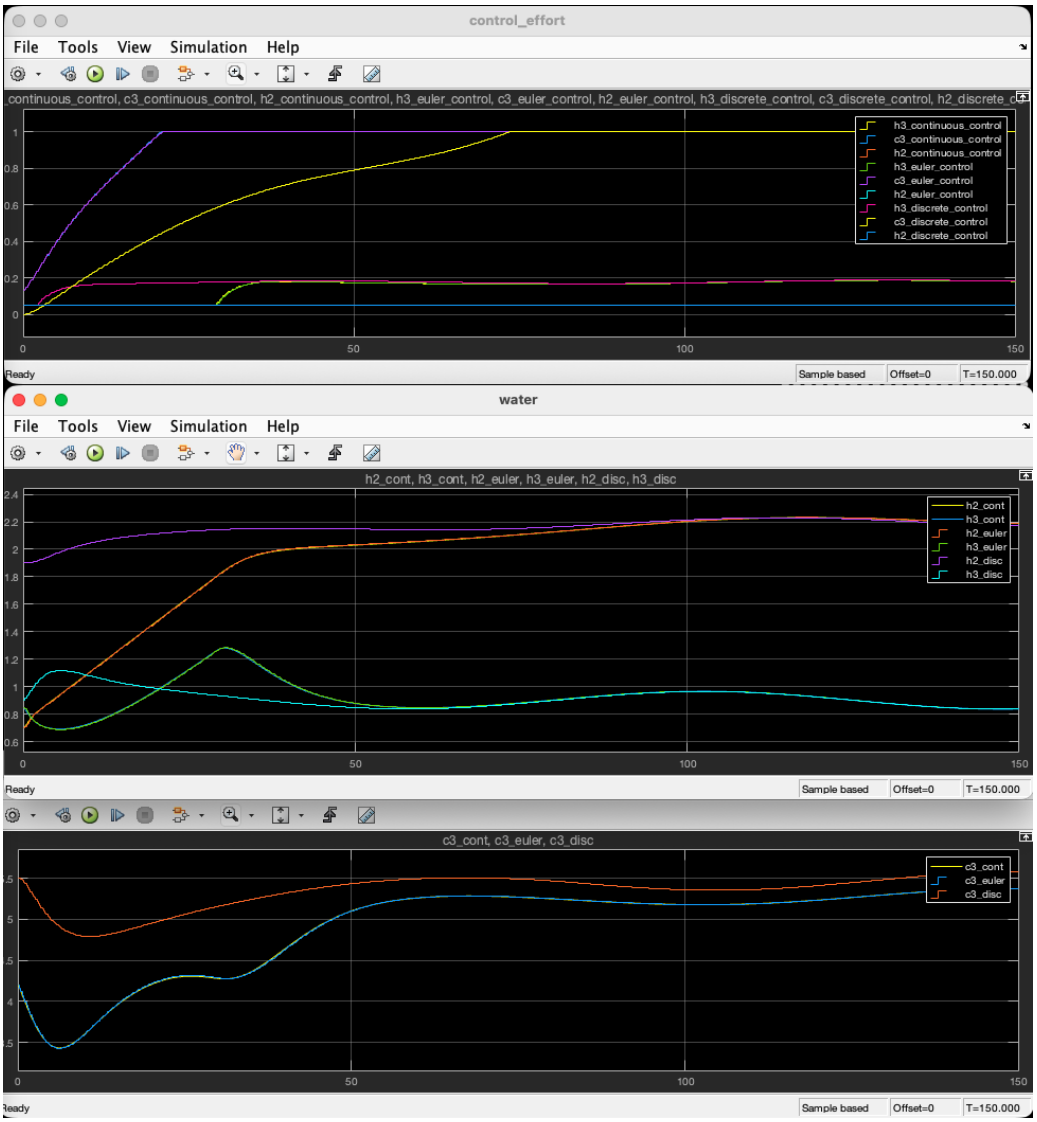


Figure 9: Three-model comparison (Task 6): nonlinear continuous vs. Euler discrete vs. linear discrete — $h_2(t)$, $h_3(t)$, $c_3(t)$ at $T_s = 0.5$ s.

9 Effect of Sampling Time (Task 7)

9.1 Analysis Methodology

Task 7 investigates how the sampling period T_s affects closed-loop performance and numerical accuracy. The same three-branch Simulink scheme from Task 6 is re-run for $T_s \in \{0.2, 0.5, 1.0, 2.0\}$ s. The analysis evaluates: (1) tracking accuracy and phase lag, (2) transient settling, (3) presence of oscillations or chattering, (4) control effort magnitude, and (5) agreement between the three model branches.

9.2 $T_s = 0.2$ s — Near-Continuous Performance

At $T_s = 0.2$ s, the discrete controller behaves very similarly to its continuous counterpart. The ZOH delay ($\approx T_s/2 \approx 0.1$ s) is negligible relative to the reference periods of 60–90 s. As shown in Fig. 10, the Euler plant model overlaps perfectly with the continuous model on all output channels. Control signals are smooth and do not exhibit visible staircase artefacts. From simulation results, this configuration represents the highest accuracy achievable within the

discrete paradigm, at the cost of a 2.5× higher controller update rate compared to $T_s = 0.5$ s.

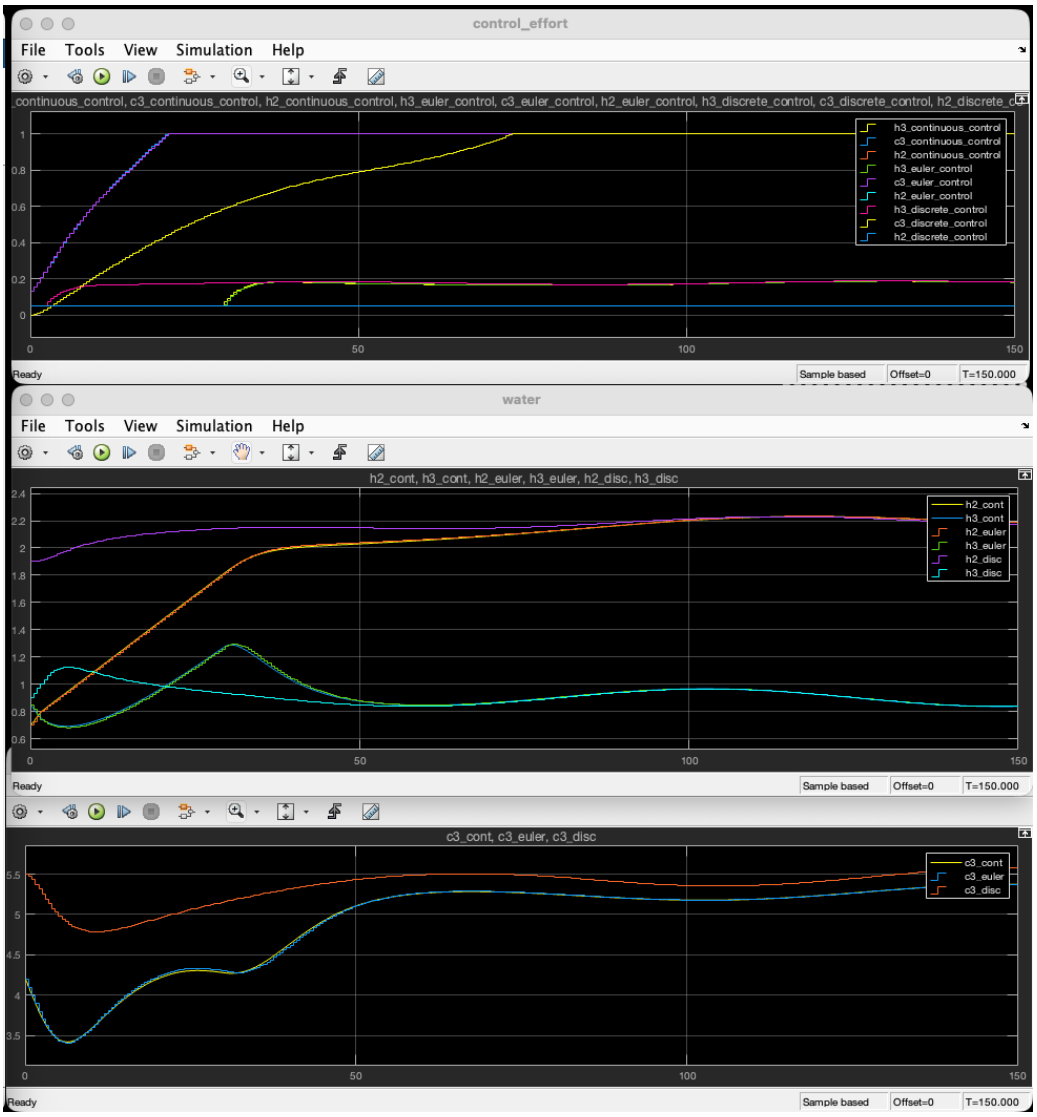


Figure 10: Sampling time analysis: $T_s = 0.2$ s — $h_2(t)$, $h_3(t)$, $c_3(t)$ tracking and control inputs.

9.3 $T_s = 0.5$ s — Recommended Operating Point

At $T_s = 0.5$ s, tracking performance remains excellent (Fig. 11). The slight phase lag — roughly 0.5 s on the h_3 sinusoid — is negligible against a reference period of 90 s. The Euler and continuous models remain nearly indistinguishable. In practice, this sampling period represents the optimal engineering trade-off: it reduces the required controller update frequency by 2.5× compared to $T_s = 0.2$ s while introducing only marginal performance degradation. This is why $T_s = 0.5$ s was adopted as the baseline for all previous tasks.

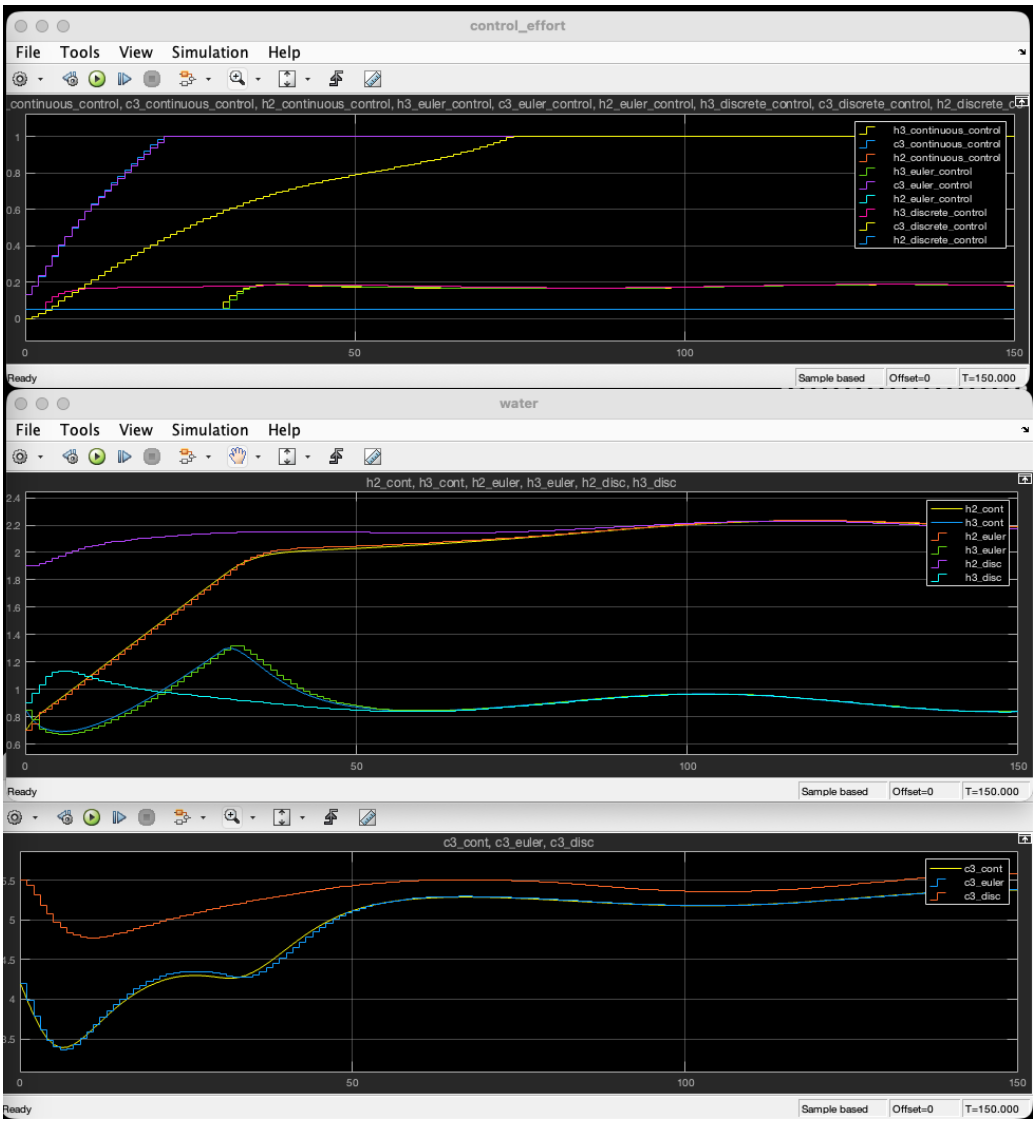


Figure 11: Sampling time analysis: $T_s = 0.5$ s — $h_2(t)$, $h_3(t)$, $c_3(t)$ tracking and control inputs.

9.4 $T_s = 1.0$ s — Visible Degradation

At $T_s = 1.0$ s, discretization effects become clearly perceptible (Fig. 12). A measurable phase lag appears in the sinusoidal tracking of h_2 and h_3. The control signals exhibit visible step increments at each sampling instant, which would increase mechanical wear on pumps and aerators in a real installation. The Euler model begins to deviate slightly from the continuous reference in the c_3 channel, which has faster dynamics ($\lambda_3 = 0.05$ s^{-1}). In practice, closed-loop stability is maintained, but the performance would likely not satisfy industrial tracking specifications.

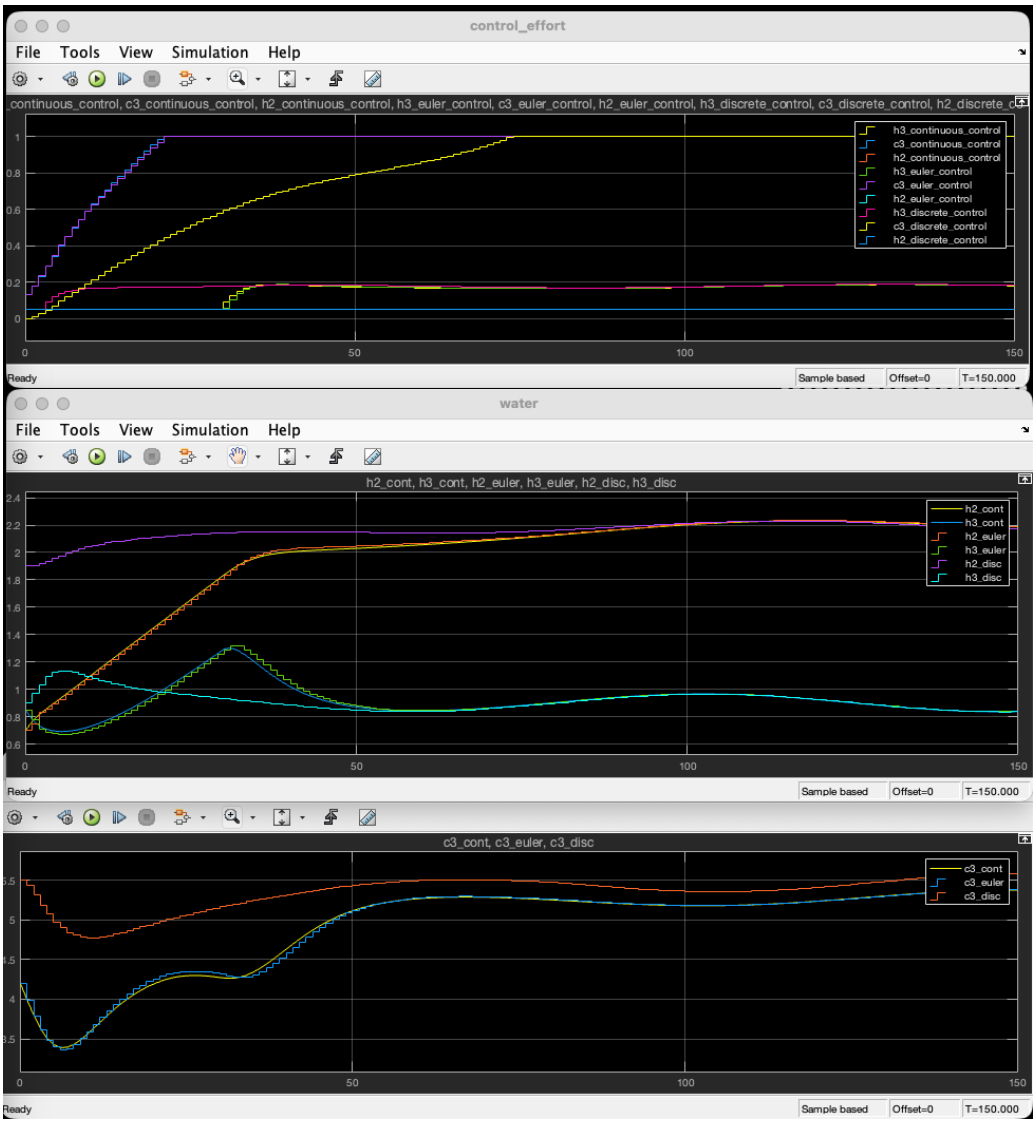


Figure 12: Sampling time analysis: $T_s = 1.0$ s — $h_2(t)$, $h_3(t)$, $c_3(t)$ tracking and control inputs.

9.5 $T_s = 2.0$ s — Significant Performance Degradation

At $T_s = 2.0$ s, performance degrades noticeably compared to smaller sampling periods (Fig. 13). Tracking errors are larger and the sinusoidal reference following for h_3 and h_2 shows a clearly perceptible phase lag. The ZOH introduces an effective delay of approximately $T_s/2 = 1.0$ s, which is no longer negligible relative to the reference signal periods. The discretization error accumulated at each Euler step is substantially larger than at $T_s = 1.0$ s, leading to a rougher, less accurate trajectory. The overall behaviour is qualitatively similar to the $T_s = 1.0$ s case, but with more pronounced degradation in all metrics: higher phase lag, larger steady-state offset, and more visible staircase artefacts in the control signals. The system remains stable, but performance is significantly degraded and would not meet reasonable tracking specifications in an industrial application. For this reason, $T_s = 2.0$ s should be avoided in practice, even though it does not cause closed-loop instability.

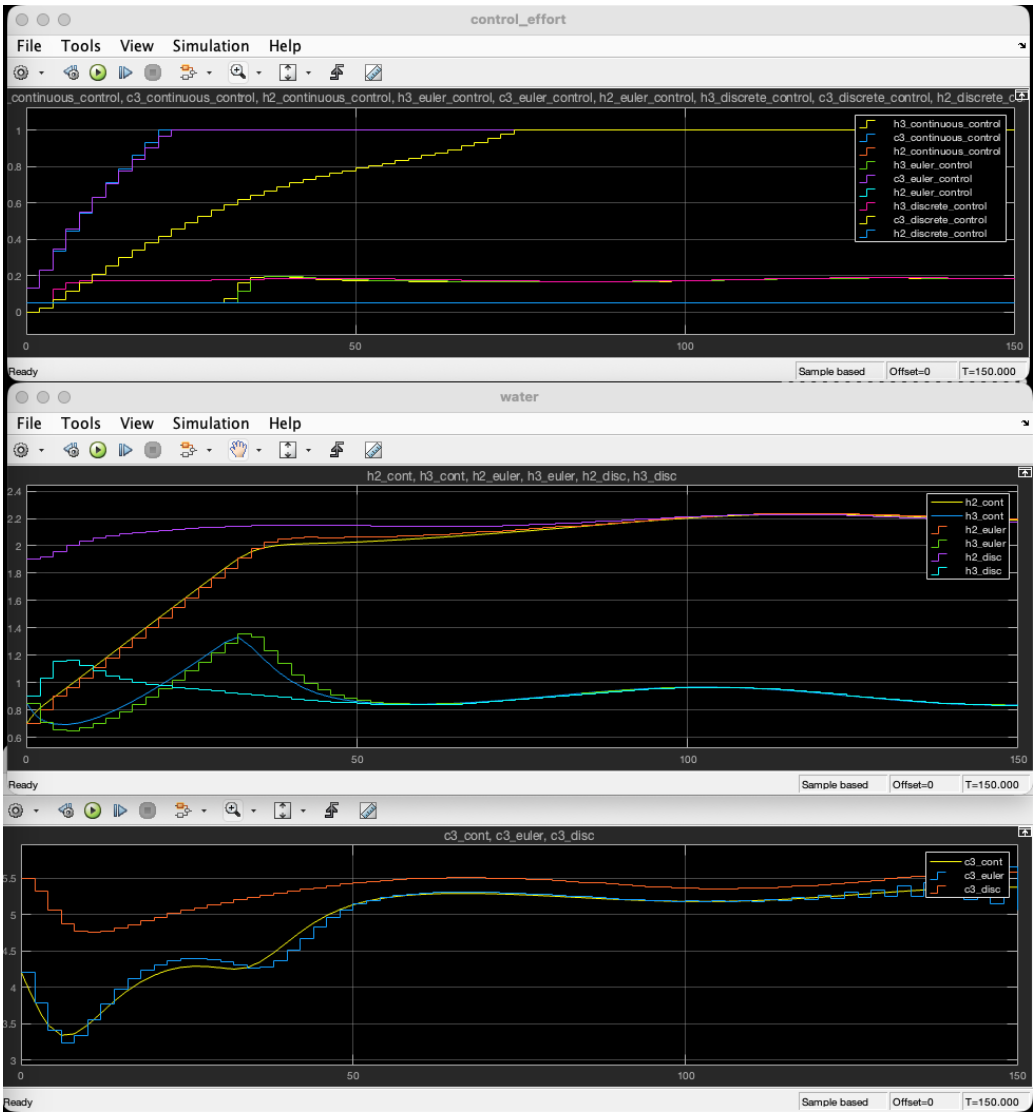


Figure 13: Sampling time analysis: $T_s = 2.0$ s — $h_2(t)$, $h_3(t)$, $c_3(t)$ tracking and control inputs.

9.6 Summary Table

Table 3 summarises the qualitative performance assessment for each sampling period.

Table 3: Qualitative performance assessment for $T_s \in \{0.2, 0.5, 1.0, 2.0\}$ s.

T_s (s)	Tracking	Phase lag	Oscillations	Euler stability
0.2	Excellent	Negligible	None	Perfect
0.5	Good	Minimal	Negligible	Very good
1.0	Moderate	Perceptible	Visible	Acceptable
2.0	Poor	High	Moderate	Stable but inaccurate

10 Bonus Observation on Disturbed Inlet Flow q_{in}

To further evaluate the robustness of the control strategy, an additional experiment was conducted by introducing a disturbance on the inlet flow q_{in}. The disturbance was modeled as a combination of a step variation and a sinusoidal component, representing both sudden and

continuous perturbations.

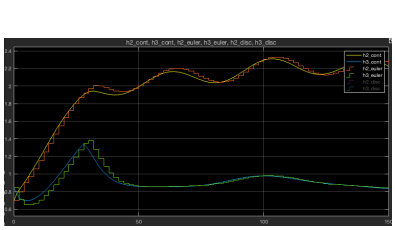
The results show that the hydraulic states (h_2, h_3) remain bounded and continue to track their references with acceptable performance, although with increased oscillations. In contrast, the oxygen dynamics (particularly c_3) exhibit significant oscillations and, in the discretized case, a tendency toward divergence under aggressive disturbance conditions.

This behavior can be explained by the combined effects of actuator saturation and integral action. In particular, the aeration input u_a frequently reaches its saturation limit, preventing the controller from applying the required corrective action. As a result, the integral term accumulates error (windup), leading to amplified oscillations in the oxygen concentration.

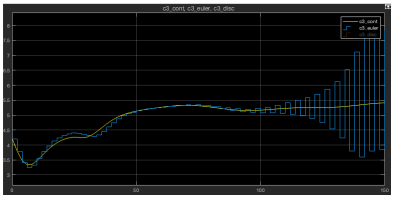
Furthermore, the nonlinear structure of the oxygen dynamics, which includes multiplicative terms and sensitivity to the hydraulic states, increases the system's susceptibility to disturbances. The discretized model (especially Euler-based) further amplifies these effects due to numerical approximation errors.

Overall, this experiment highlights the limitations of the control scheme under strong disturbances and confirms that, while the system is robust in nominal conditions, its performance degrades when operating far from the equilibrium point.

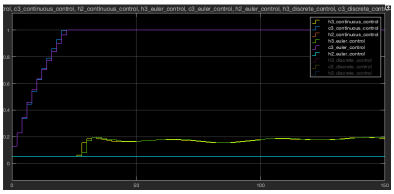
Disturbance Response Results



(a) Water levels response



(b) Oxygen concentration response



(c) Control effort

Figure 14: System response under disturbed inlet flow q_{in}. The disturbance induces oscillations in all states, with a particularly strong impact on the oxygen dynamics.

Effect of Controller Gain Variation

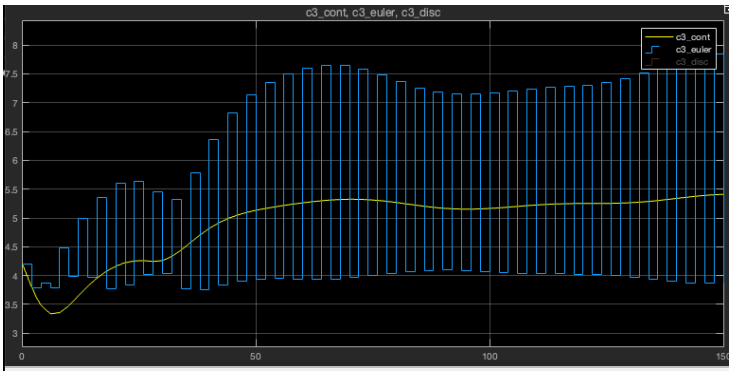


Figure 15: Effect of increased proportional gain on oxygen concentration c_3. Higher gain leads to amplified oscillations and potential instability due to saturation and windup effects.

11 Conclusions

This work presented a complete modeling and digital control study of a cascaded three-tank aeration system. The nonlinear continuous-time model, derived from Torricelli hydraulic laws and oxygen mass balances, accurately captures the coupled level-concentration dynamics of the process.

Three independent PI controllers were designed in continuous time and successfully extended to discrete time. Both the ZOH hybrid architecture (Task 3) and the fully Euler-discretized plant (Task 4) demonstrated stable and accurate tracking for sampling periods $T_s \leq 0.5$ s. The physical cascade constraint ($h_2 > h_3$) imposes a structural limitation on the achievable range of h_2, which must be acknowledged in any realistic control specification.

The equilibrium point was analytically derived from the steady-state balance equations, confirming that a hydraulic equilibrium requires $u_d^* = q_{\text{in}}$. Symbolic Jacobian computation via MATLAB's Symbolic Math Toolbox provided exact linearization matrices (A, B, C, D) without numerical approximation errors. The ZOH-discretized linear model proved to be a reliable representation of the system for control purposes in the neighbourhood of the chosen operating point.

The three-branch comparison (Task 6) confirmed that, at $T_s = 0.5$ s, the Euler discrete model and the continuous reference are practically indistinguishable, while the linear discrete model provides a satisfactory but slightly less accurate approximation due to the inherent limitations of local linearization.

The sampling time analysis (Task 7) identified $T_s = 0.5$ s as the recommended design choice: it provides near-continuous tracking performance while halving the computation rate with respect to $T_s = 0.2$ s. Values above 1.0 s lead to noticeable performance degradation. $T_s = 2.0$ s results in significant deterioration of tracking quality — increased phase lag, larger discretization error, and rougher control signals — and should be avoided in practice, although the system remains stable.